RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Academic Year 2023 – 2024 (Odd Semester)

Name of the Faculty	:	Dr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Course Code and Title	:	MF8071 Additive Manufacturing
Year and Branch	:	IV-Mechanical
Unit/ Topic	:	I – INTRODUCTION – AM Process Chain, Classification.
Course Outcome	:	Acquire knowledge about a working principle and construction of Additive Manufacturing technologies, their potential to support design and manufacturing.
Topic Learning Outcome	:	Gain a comprehensive understanding of Additive Manufacturing (AM) process chains and classifications, including the working principles, construction, and the potential of AM technologies to enhance design and manufacturing capabilities.
Activity Chosen		Exploring the Foundations: NPTEL's Introductory Lecture
Time Allotted for Activity	:	50 minutes
Date of the activity	:	28.07.2023 (Friday) – 9.00 AM to 9.50 AM
No.of Students participated	:	28
Justification	:	Engaging with the NPTEL video on the Fundamentals of Additive Manufacturing Technologies, specifically Week 1 Lecture 1, Introduction to Additive Manufacturing by Prof. Sajan Kapil from the Mechanical Engineering department at IIT Guwahati, offers a crucial foundation for comprehending the working principles and construction of Additive Manufacturing (AM) technologies. This activity serves as a strategic initiative to deepen understanding

		about AM processes and their classifications, providing insights into the transformative potential of these technologies in supporting both design and manufacturing domains. By tapping into authoritative academic resources, participants can gain valuable theoretical knowledge that lays the groundwork for informed exploration and application of AM in various industries, fostering a holistic understanding of this cutting-edge technology.
Details of Implementation	:	The implementation of the topic "AM Process Chain, Classification" involves organizing a comprehensive session based on the NPTEL video "Fundamentals of Additive Manufacturing Technologies" by Prof. Sajan Kapil from IIT Guwahati. The session is scheduled for 50 minutes on 28.07.2023 (Friday) from 9.00 AM to 9.50 AM. **Details of Implementation:** 1. **Introduction (5 minutes):** Start with a brief overview of the importance of Additive Manufacturing (AM) in modern design and manufacturing processes. Highlight its advantages and how it complements traditional manufacturing methods. 2. **Video Lecture (30 minutes):** Play the NPTEL video by Prof. Sajan Kapil. Ensure a conducive environment for participants to absorb key concepts, including the working principles and construction of various AM technologies. Emphasize the potential of AM in supporting innovative design approaches and efficient manufacturing processes. 3. **Q&A Session (10 minutes):** Facilitate a question-and-answer session to address any queries or clarifications from the participants

		regarding the content of the lecture. Encourage discussions on the real-world applications and challenges associated with implementing AM technologies. 4. **Class Activity (5 minutes):** Conclude with a short interactive activity, such as a brief discussion on specific AM case studies or examples. This reinforces key takeaways from the lecture and allows participants to apply their newfound knowledge.

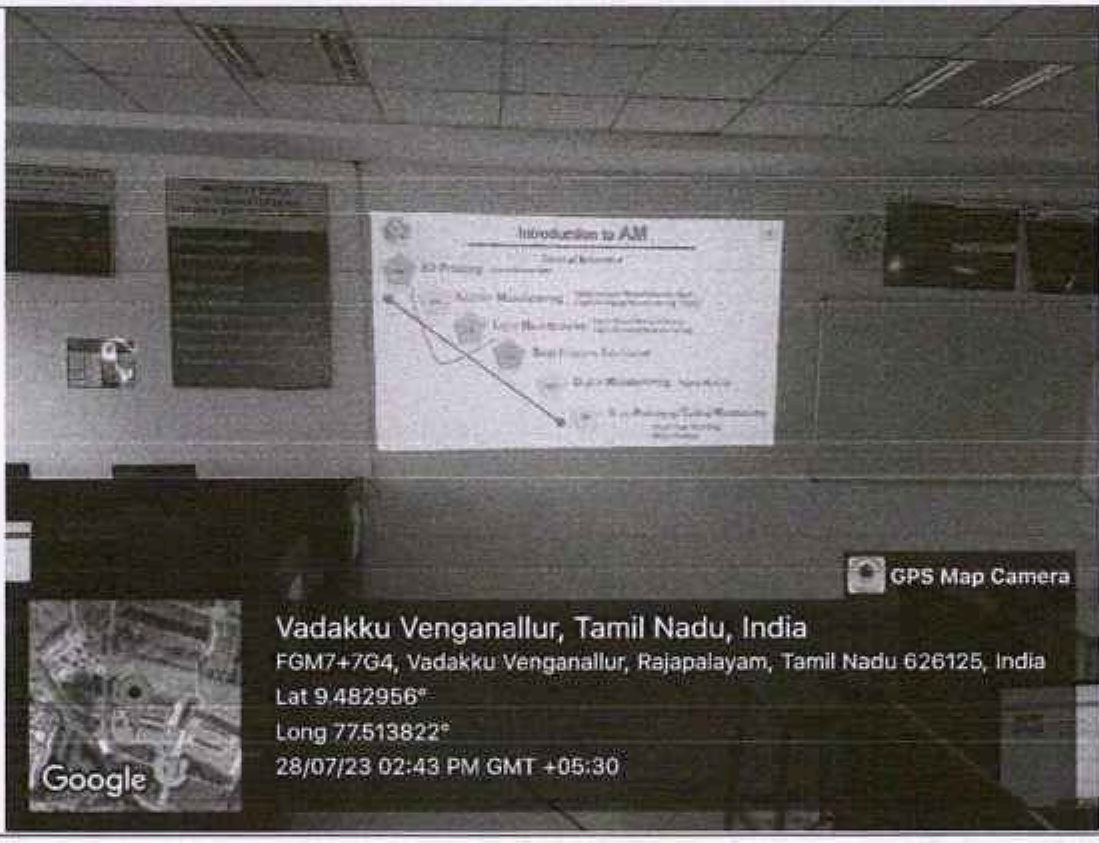
PO/PSO mapping for the activity

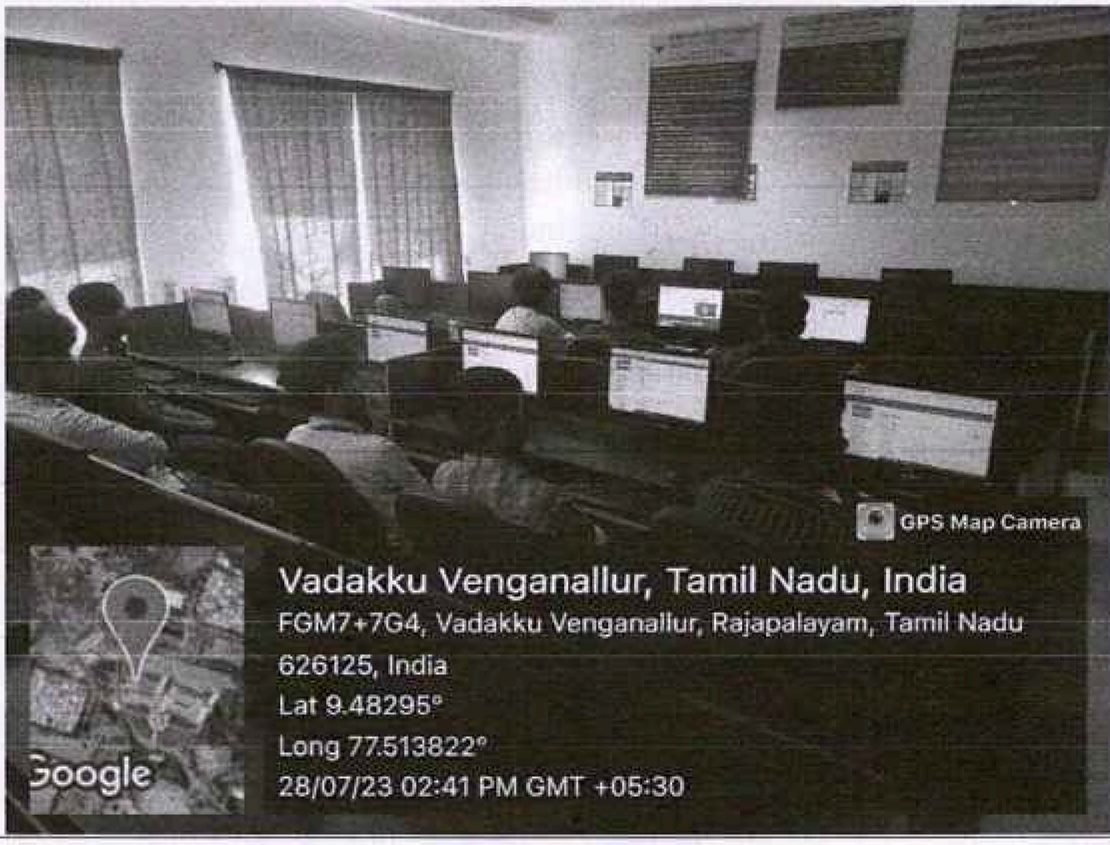
Course Outcome /Programme Outcome	PO1	PO4	PO5	PO7	PO12	PSO1
Level of Mapping	3	2	3	2	1	1
Justification for correlation	Attend NPTEL lecture on "Introduction to Additive Manufacturing" to acquire understanding of AM technologies' working principles and	Analyze the content of the NPTEL lecture, specifically focusing on the fundamental aspects of Additive Manufacturing	To identify modern engineering tools such as PTC CREO software for engineering activities	Consider the environmental and sustainability aspects associated with Additive Manufacturing processes as discussed	Recognize the importance of continuous learning in staying updated on advancements in Additive Manufacturing technologies and	To design component using CAD package

	constructions.	technologies.		in the lecture.	their applications.	

(1 – Low, 2- Moderate, 3-High)

Activity Glimpses





Reflective Critique:

i. Feedback of practice from students and other stakeholders:

Students gave positive oral feedback

ii. Benefit of the practice:

Gain in-depth insights into the working principles, construction, and potential design and manufacturing support offered by Additive Manufacturing technologies through a focused NPTEL lecture, enhancing knowledge in this rapidly evolving field.

iii. Whether the practice is adopted in any of the courses early:

Yes, in the same course

Challenges faced in implementation:

Coordinating a comprehensive exploration of Additive Manufacturing (AM) within a limited 50-minute NPTEL lecture presents challenges in balancing theoretical depth with practical insights. Addressing the working principles and constructions of various AM technologies, as well as their implications for design and manufacturing, demands a focused and streamlined approach. Efficiently covering this breadth of content requires strategic selection of key concepts and effective communication by Prof. Sajan Kapil to maximize student understanding within the constrained time frame.

References: Nil

CO1: Students will be able to Acquire knowledge about a working principle and construction of Additive Manufacturing technologies, their potential to support design and manufacturing

15/11/23

Faculty In Charge

15/11/23

HoD/Mech